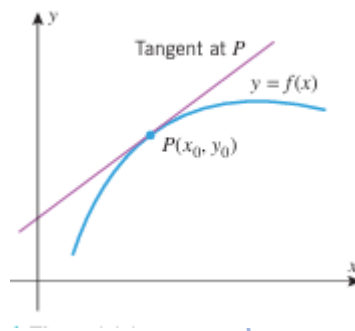
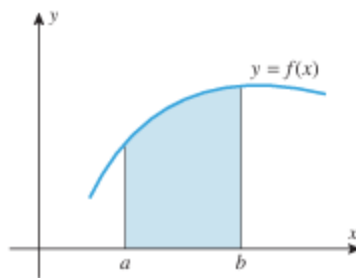


LIMITS & CONTINUITY

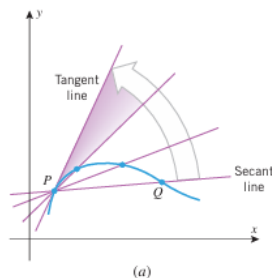
The Tangent Line Problem: Given a function f and a point $P(x_0, y_0)$ on its graph, find an equation of the line i.e. tangent to the graph at p or the tangent line problem is called differential Calculus.



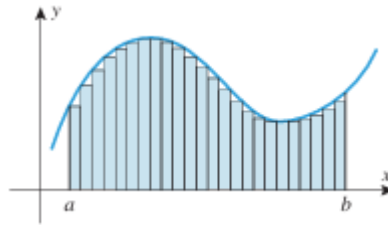
The Area Problem: Given a function f find the area between the graph of f and an interval $[a, b]$ on the x -axis. or the area problem is called an integral calculus.



Tangent Line and Limits: Suppose the tangent line at a point P on a Curve in the xy -plane and that Q is a point that lies on the curve and is different from p . The line through P and Q is called a Secant line for the Curve at P . If we move the point Q along the curve toward P , then the secant line will rotate toward a limiting position. That line in the limiting position is considered as tangent line at p or "The limit of the slopes of secant lines as the two points on the secant line get closer together.



Area and Limits: The limit part for area in calculus is approximating the area under a curve by rectangles. Hence, the sum of the area of the rectangles become the area under the curve.



Limits: It describe how a function behaves as the independent variable approaches a given Value.

For e.g. $f(x) = x^2 - x + 1$ for $\lim_{x \rightarrow 2}$

$$\lim_{x \rightarrow 2} (x^2 - x + 1) = 3$$

It $\lim_{x \rightarrow 2} (x^2 - x + 1)$ is 3 as x approaches 2 from either side.

Limits (In formal): If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but not equal to a) then:

$$\lim_{x \rightarrow a} f(x) = L$$

which is "the limit of $f(x)$ as x approaches a is L ."

One Side Limits: If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a :

i) but greater than a then; $\lim_{x \rightarrow a+} f(x) = L$

ii) but less than a then; $\lim_{x \rightarrow a-} f(x) = L$

The Relationship between one-sided limits and two sided limits: The two sided limit of a function $f(x)$ exists at a iff both of the one sided limit exists at a and have the same values i.e.

$$\lim_{x \rightarrow a} f(x) = L \text{ iff } \lim_{x \rightarrow a+} f(x) = L = \lim_{x \rightarrow a-} f(x)$$

Infinite Limits: The expressions

$$\lim_{x \rightarrow a+} f(x) = +\infty \text{ and } \lim_{x \rightarrow a-} f(x) = +\infty$$

denoted that $f(x)$ increases without bound as x approaches to a from the left and the right respectively, If both are true then:

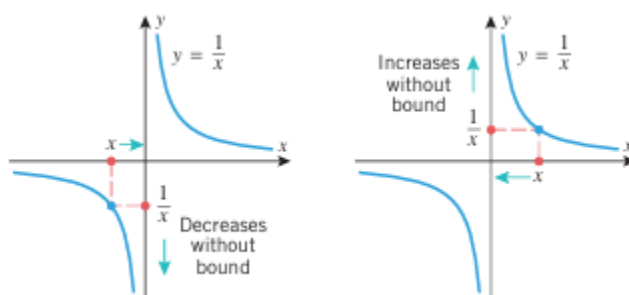
$$\lim_{x \rightarrow a} f(x) = +\infty$$

Similarly;

$$\lim_{x \rightarrow a^+} f(x) = -\infty \text{ and } \lim_{x \rightarrow a^-} f(x) = -\infty$$

denoted that $f(x)$ decreases without bound as x approaches to a from the left and the right respectively, If both are true then:

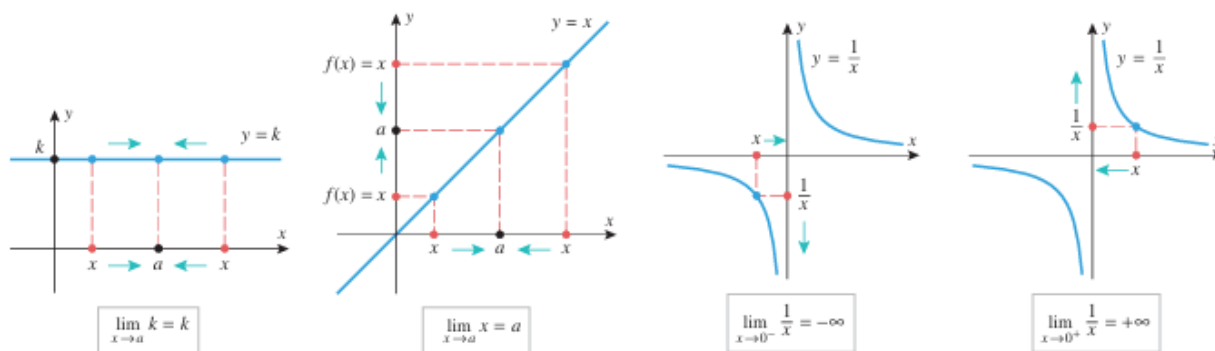
$$\lim_{x \rightarrow a} f(x) = -\infty$$



Computing Limits:

Theorem: Let a and k be real numbers.

$$\text{a. } \lim_{x \rightarrow a} K = K \quad \text{b. } \lim_{x \rightarrow a} x = a \quad \text{c. } \lim_{x \rightarrow 0^-} 1/x = -\infty \quad \text{d. } \lim_{x \rightarrow 0^+} 1/x = +\infty$$



Theorem: Let a be a real number, and Suppose that:

$$\lim_{x \rightarrow a} f(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = L_2$$

i.e. the limits exist and have values L_1 and L_2 respectively, then:

$$\text{a. } \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L_1 + L_2$$

$$\text{b. } \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L_1 - L_2$$

$$\text{c. } \lim_{x \rightarrow a} [f(x)g(x)] = (\lim_{x \rightarrow a} f(x))(\lim_{x \rightarrow a} g(x)) = L_1 L_2$$

$$d. \lim_{x \rightarrow a} [f(x)/g(x)] = (\lim_{x \rightarrow a} f(x)) / (\lim_{x \rightarrow a} g(x)) = L_1 / L_2 \quad \forall L_2 \neq 0$$

$$e. \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L_1} \quad \forall L_1 > 0 \text{ if } n \text{ is even.}$$

Also, these statements are true for the one- sided limits as $x \rightarrow a^+$ or as $x \rightarrow a^-$

Theorem: for any polynomial

$$P(X) = C_0 + C_1X + \dots + C_n X^n$$

and any real number a,

$$\lim_{x \rightarrow a} P(X) = C_0 + C_1X + \dots + C_n X^n = P(a)$$

Theorem: for any rational function

$$\text{Let } f(x) = \frac{p(x)}{q(x)}$$

be a rational function and let a be any real number

- If $q(a) \neq 0$, then $\lim_{x \rightarrow a} f(x) = f(a)$
- If $q(a) = 0$, $p(a) \neq 0$ then $\lim_{x \rightarrow a} f(x)$ does not exist

Limits of piecewise – Defined Function:

To find the limit of a piece wise function, you can calculate the the one-sided limits separately for each interval of the function. It has different formulas for different ranges of values.

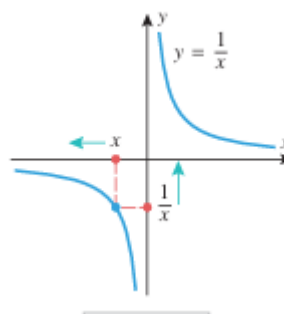
Limits at Infinity: If the values of $f(x)$ eventually get as close as we like to a number L

- as x increases without bound, then we write

$$\lim_{x \rightarrow +\infty} f(x) = L$$

- as x decreases without bound, then we write

$$\lim_{x \rightarrow -\infty} f(x) = L$$



Limit Laws for Limits at Infinity:

- $\lim_{x \rightarrow +\infty} (f(x))^n = (\lim_{x \rightarrow +\infty} f(x))^n$ or $\lim_{x \rightarrow -\infty} (f(x))^n = (\lim_{x \rightarrow -\infty} f(x))^n$
- $\lim_{x \rightarrow +\infty} kf(x) = k \lim_{x \rightarrow +\infty} f(x)$ or $\lim_{x \rightarrow -\infty} kf(x) = k \lim_{x \rightarrow -\infty} f(x)$
- $\lim_{x \rightarrow +\infty} K = K$ or $\lim_{x \rightarrow -\infty} K = K$

Infinite Limits at infinity:

If the values of $f(x)$ eventually get as close as we like to a number L

- i) as x increases without bound, as $x \rightarrow +\infty$ or $x \rightarrow -\infty$

$$\lim_{x \rightarrow +\infty} f(x) = +\infty \text{ or } \lim_{x \rightarrow -\infty} f(x) = +\infty$$

- ii) as x decreases without bound, then we write

$$\lim_{x \rightarrow +\infty} f(x) = -\infty \text{ or } \lim_{x \rightarrow -\infty} f(x) = -\infty$$

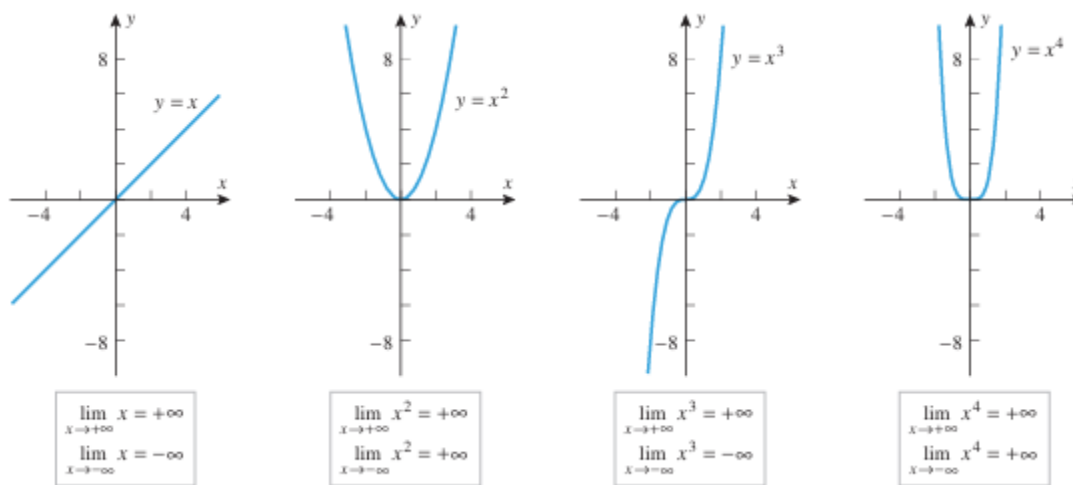
Limits Of x^n as $x \rightarrow \pm\infty$: The end behavior of the polynomials x^n for $n=1,2,3$ and 4

Special cases are:

$$\lim_{x \rightarrow +\infty} x^n = +\infty \quad n=1,2,3,\dots$$

$$\lim_{x \rightarrow -\infty} x^n = \begin{cases} -\infty & n = 1,3,5, \dots \\ +\infty & n = 2,4,6, \dots \end{cases}$$

Multiplying x^n by a positive real number does not affect limits but negative real number reverse the sign.



Limits Of Polynomial as $x \rightarrow \pm\infty$:

The end behavior of a polynomial matches the end behavior of its highest degree term if $C_n \neq 0$ then:

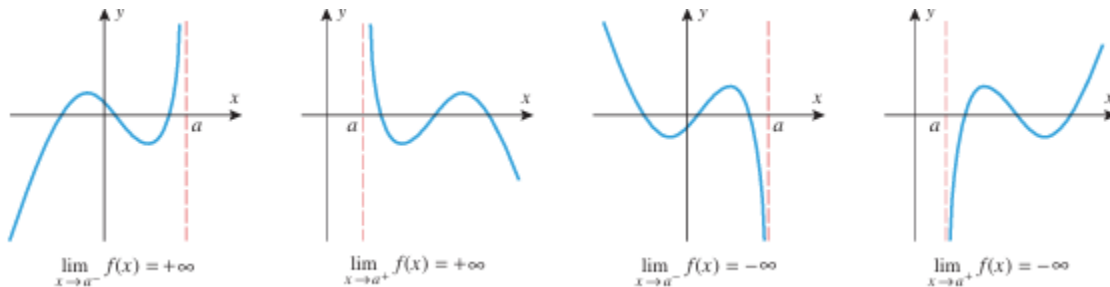
$$\lim_{x \rightarrow \pm\infty} P(X) = C_0 + C_1X + \dots + C_nX^n = \lim_{x \rightarrow \pm\infty} C_nX^n$$

Limits Of Rational functions as $x \rightarrow \pm\infty$: The end behavior of a rational function matches the end behavior of the quotient of the highest degree term in the numerator divided by the highest degree term in the denominator.

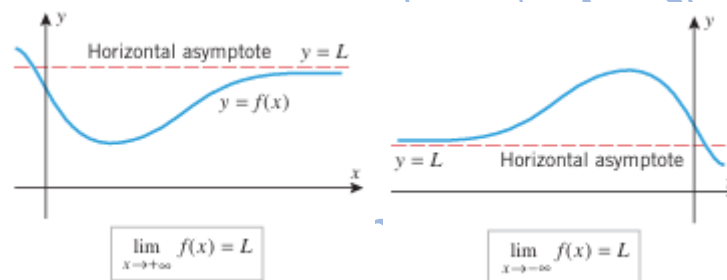
Limits Involving Radicals:

It can be computed by taking ratios of square root functions and its horizontal asymptotes.

Vertical Asymptote: It is a vertical line on a graph that a function can't touch. It occurs when function is undefined at a specific value of x .



Horizontal Asymptote: It is a horizontal line that a function approaches but never crosses. It occurs when a function is undefined at a specific value of x .

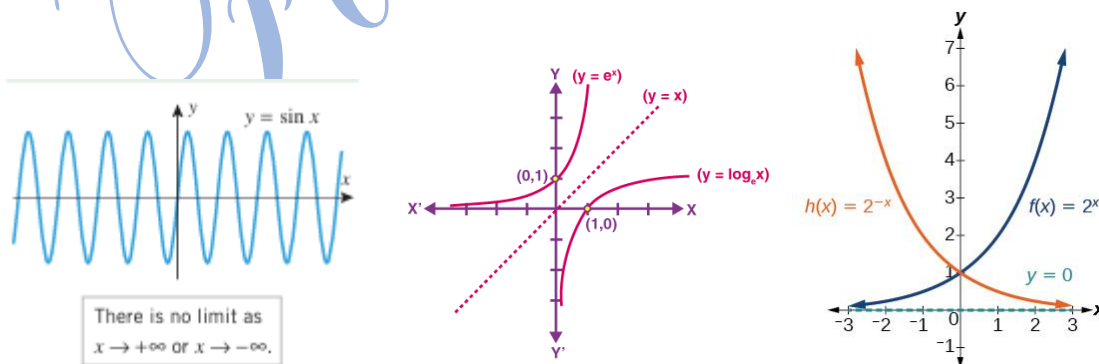


End behavior of Trigonometric, Exponential, and Logarithmic:

In general, the trigonometric functions fail to have limits as $x \rightarrow \pm\infty$ because of periodicity.

In general, the functions e^x and $\ln x$ both increase without bound as $x \rightarrow +\infty$. We have:

1. $\lim_{x \rightarrow -\infty} \ln x = -\infty$ 2. $\lim_{x \rightarrow +\infty} e^x = +\infty$ 3. $\lim_{x \rightarrow -\infty} e^x = 0$ 4. $\lim_{x \rightarrow 0^+} \ln x = -\infty$

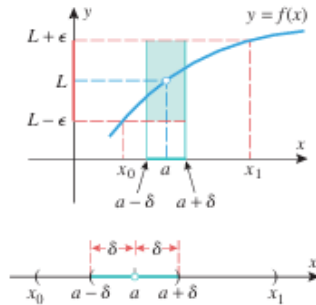


Limit Definition: let $f(x)$ be defined for all x in some open interval containing the number a with the possible exception that $f(x)$ need not be defined at a . i.e.

$$\lim_{x \rightarrow a} f(x) = L$$

If given any number $\epsilon > 0$ we can find a number $\delta > 0$ such that

$$|f(x) - L| < \epsilon \text{ if } 0 < |x - a| < \delta$$



The value of δ is not unique: in definition of limit the value of δ is not unique, once a value of δ that fulfills the requirements of the definition then any smaller +ve numbers δ_1 will also fulfill those requirements. i.e. if it is true that

$$|f(x) - L| < \epsilon \text{ if } 0 < |x - a| < \delta$$

Then it will also be true that

$$|f(x) - L| < \epsilon \text{ if } 0 < |x - a| < \delta_1$$

That is if:

$$|f(x) - L| < \epsilon$$

Is satisfied for all x in the largest set then it will be satisfied for all x in the subset.

Limits as $x \rightarrow \pm\infty$:

- i. Let $f(x)$ be defined for all x in some infinite open interval extending in the positive x -direction i.e.

$$\lim_{x \rightarrow +\infty} f(x) = L$$

If any number $\epsilon > 0$, there corresponds a +ve number N such that

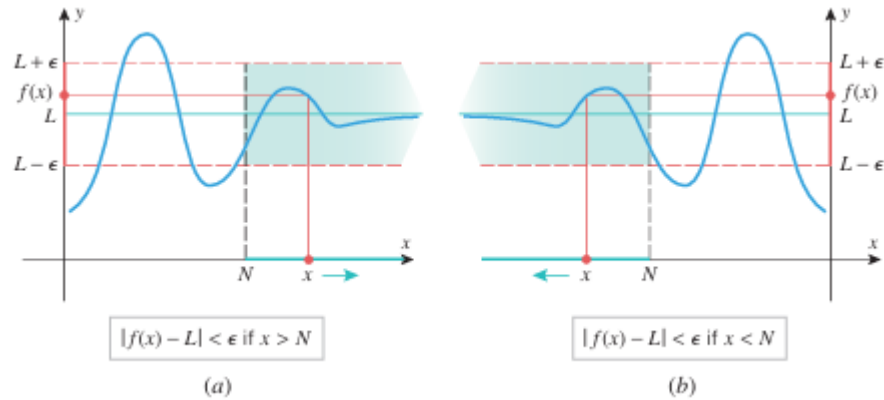
$$|f(x) - L| < \epsilon \text{ if } x > N$$

- ii. Let $f(x)$ be defined for all x in some infinite open interval extending in the negative x -direction i.e.

$$\lim_{x \rightarrow -\infty} f(x) = L$$

If any number $\epsilon > 0$, there corresponds a -ve number N such that

$$|f(x) - L| < \epsilon \text{ if } x < N$$



Infinite Limits as $x \rightarrow \pm\infty$:

- i. Let $f(x)$ be defined for all x in some open interval containing a except that $f(x)$ need not to be defined at a i.e.

$$\lim_{x \rightarrow a} f(x) = +\infty$$

If given any +ve number M , number we can find a number $\delta > 0$ such that $f(x)$ satisfies

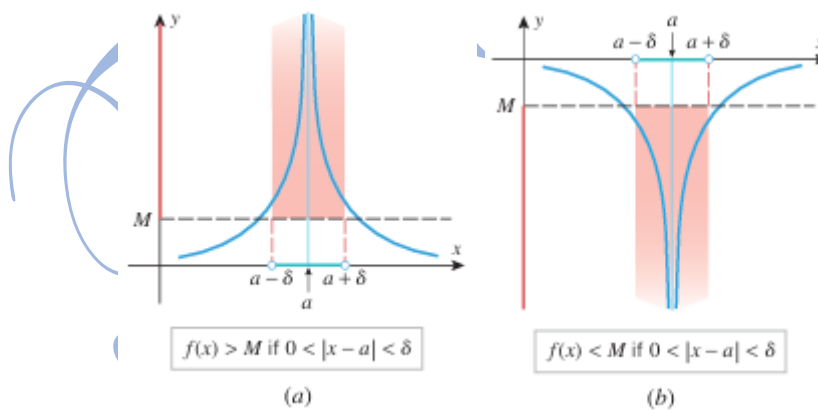
$$f(x) > M \text{ if } 0 < |x - a| < \delta$$

- ii. Let $f(x)$ be defined for all x in some open interval containing a except that $f(x)$ need not to be defined at a i.e.

$$\lim_{x \rightarrow a} f(x) = -\infty$$

If given any +ve number M , number we can find a number $\delta > 0$ such that $f(x)$ satisfies

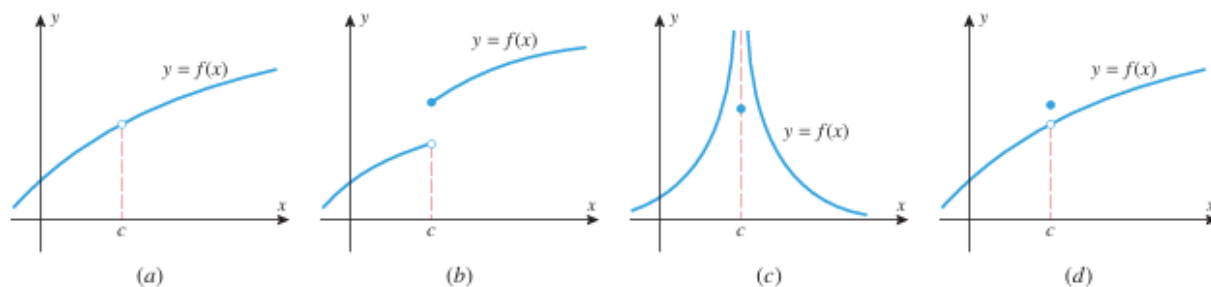
$$f(x) < -M \text{ if } 0 < |x - a| < \delta$$



Continuity:

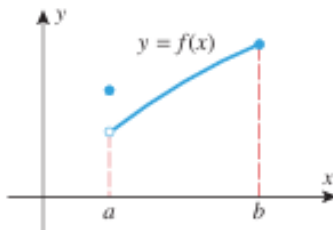
A function f is said to be continuous at $x=c$ provided the following conditions are satisfied:

- i. $f(c)$ is defined
- ii. $\lim_{x \rightarrow c} f(x)$ exists
- iii. $\lim_{x \rightarrow c} f(x) = f(c)$



Continuity on an Interval:

- i. A function is said to be continuous on a closed interval $[a, b]$ if the following conditions on (a, b) , $[a, +\infty)$, $(-\infty, b]$, (a, b) and $[a, b]$.
- ii. f is continuous from the right at a .
- iii. f is continuous from the left at b .



Some properties of continuous function:

If a functions f and g are continuous at c , then

- a. $f + g$ is continuous at c .
- b. $f - g$ is continuous at c .
- c. $f \cdot g$ is continuous at c .
- d. f/g is continuous at c , iff $g(c) \neq 0$ and has a discontinuity at c if $g(c) = 0$

Continuity of Polynomials and Rational Functions:

- a. A polynomial is continuous every where i.e. $\lim_{x \rightarrow a} p(x) = p(a)$
- b. A rational function is continuous at every point where the denominator is non-zero, and has discontinuities at the points where the denominator is zero.

Continuity of Compositions:

- i. If $\lim_{x \rightarrow c} g(x) = L$ and if the function f is continuous at L , then

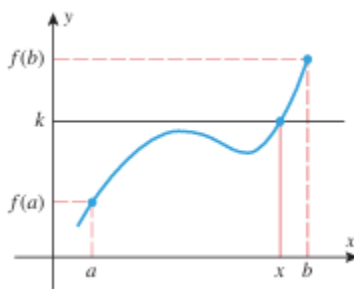
$$\lim_{x \rightarrow c} f(g(x)) = f \lim_{x \rightarrow c} (g(x))$$

This equality remains valid if $\lim$ is replaced everywhere by one of $\lim_{x \rightarrow c+}$, $\lim_{x \rightarrow c-}$, $\lim_{x \rightarrow \pm\infty}$.

- ii. If the function g is continuous at c , and the function f is continuous at $g(c)$, then the composition $f \circ g$ is continuous at c .
- iii. If the function g and f is continuous everywhere, then the composition $f \circ g$ is continuous everywhere.

The Intermediate Value Theorem:

If f is continuous on a closed interval $[a, b]$ and K is any number between $f(a)$ and $f(b)$ are non-zero and have opposite signs, then there is at least one solution of the equation $f(x)=0$ in the interval (a, b) .



Continuity of Trigonometric Functions:

If c is any number in the natural domain of the stated trigonometric function, then

- i. $\lim_{x \rightarrow c} \sin x = \sin c$ ii. $\lim_{x \rightarrow c} \cos x = \cos c$ iii. $\lim_{x \rightarrow c} \tan x = \tan c$
- iv. $\lim_{x \rightarrow c} \operatorname{cosec} x = \operatorname{cosec} c$ v. $\lim_{x \rightarrow c} \sec x = \sec c$ vi. $\lim_{x \rightarrow c} \cot x = \cot c$

Continuity of Inverse Function:

If f is a one to one function that is continuous at each point of its domain, then f^{-1} is continuous at each point of its domain, i.e. f^{-1} is continuous at each point in the range of f .

Theorem: let $b > 0, b \neq 1$

- i. The function b^x is continuous on $(-\infty, +\infty)$.
- ii. The function $\log_b x$ is continuous on $(0, +\infty)$.

Theorem (the squeezing theorem):

Let f , g and h be functions satisfying

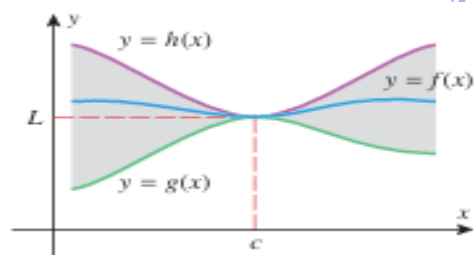
$$g(x) \leq f(x) \leq h(x)$$

for all x in some open interval containing the number c , with the possible exception that the inequalities need not hold at c . If g and h have the same limit as x approaches c , say

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$$

Then f also has this limit as x approaches c that is,

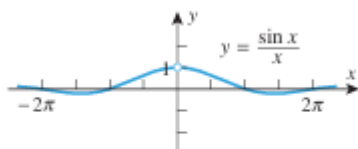
$$\lim_{x \rightarrow c} f(x) = L$$



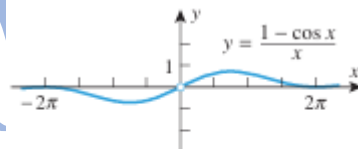
Theorem:

a. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

b. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$



$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$



$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

▲ Figure 1.6.3

1–10 In these exercises, make reasonable assumptions about the graph of the indicated function outside of the region depicted. ■

1. For the function g graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow 0^-} g(x)$ (b) $\lim_{x \rightarrow 0^+} g(x)$
 (c) $\lim_{x \rightarrow 0} g(x)$ (d) $g(0)$.

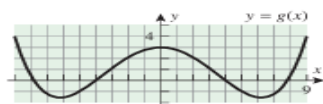


Figure Ex-1

2. For the function G graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow 0^-} G(x)$ (b) $\lim_{x \rightarrow 0^+} G(x)$
 (c) $\lim_{x \rightarrow 0} G(x)$ (d) $G(0)$.

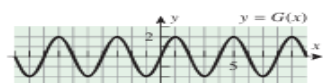


Figure Ex-2

3. For the function f graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow 3^-} f(x)$ (b) $\lim_{x \rightarrow 3^+} f(x)$
 (c) $\lim_{x \rightarrow 3} f(x)$ (d) $f(3)$.

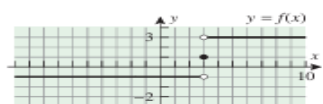


Figure Ex-3

4. For the function f graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow 2^-} f(x)$ (b) $\lim_{x \rightarrow 2^+} f(x)$
 (c) $\lim_{x \rightarrow 2} f(x)$ (d) $f(2)$.

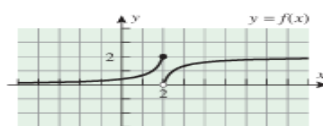


Figure Ex-4

5. For the function F graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow -2^-} F(x)$ (b) $\lim_{x \rightarrow -2^+} F(x)$
 (c) $\lim_{x \rightarrow -2} F(x)$ (d) $F(-2)$.



Figure Ex-5

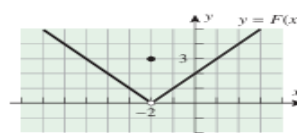


Figure Ex-5

6. For the function G graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow 0^-} G(x)$ (b) $\lim_{x \rightarrow 0^+} G(x)$
 (c) $\lim_{x \rightarrow 0} G(x)$ (d) $G(0)$.

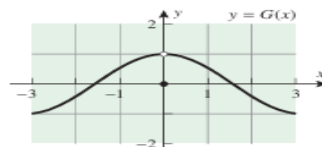


Figure Ex-6

7. For the function f graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow 3^-} f(x)$ (b) $\lim_{x \rightarrow 3^+} f(x)$
 (c) $\lim_{x \rightarrow 3} f(x)$ (d) $f(3)$.

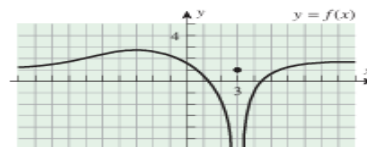


Figure Ex-7

8. For the function ϕ graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow 4^-} \phi(x)$ (b) $\lim_{x \rightarrow 4^+} \phi(x)$
 (c) $\lim_{x \rightarrow 4} \phi(x)$ (d) $\phi(4)$.

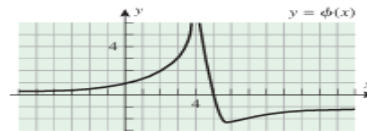


Figure Ex-8

9. For the function f graphed in the accompanying figure on the next page, find
 (a) $\lim_{x \rightarrow -2^-} f(x)$ (b) $\lim_{x \rightarrow -2^+} f(x)$
 (c) $\lim_{x \rightarrow -2} f(x)$ (d) $\lim_{x \rightarrow -2} f(x)$
 (e) $\lim_{x \rightarrow 2^+} f(x)$ (f) the vertical asymptotes of the graph of f .

60 Chapter 1 / Limits and Continuity

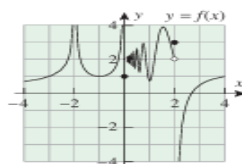


Figure Ex-9

10. For the function f graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow -2^-} f(x)$ (b) $\lim_{x \rightarrow -2^+} f(x)$ (c) $\lim_{x \rightarrow -2} f(x)$
 (d) $\lim_{x \rightarrow 0^-} f(x)$ (e) $\lim_{x \rightarrow 0^+} f(x)$ (f) $\lim_{x \rightarrow 2^+} f(x)$
 (g) the vertical asymptotes of the graph of f .

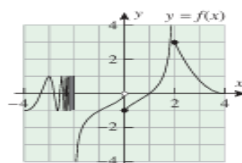


Figure Ex-10

11–12 (i) Complete the table and make a guess about the limit indicated. (ii) Confirm your conclusions about the limit by graphing a function over an appropriate interval. [Note: For the trigonometric functions, be sure to put your calculating and graphing utilities in radian mode.] ■

11. $f(x) = \frac{\sin 2x}{x}$; $\lim_{x \rightarrow 0} f(x)$

13. (a) $\lim_{x \rightarrow 1} \frac{x-1}{x^3-1}$; $x = 2, 1.5, 1.1, 1.01, 1.001, 0, 0.5, 0.9, 0.99, 0.999$
 (b) $\lim_{x \rightarrow 1^+} \frac{x+1}{x^3-1}$; $x = 2, 1.5, 1.1, 1.01, 1.001, 1.0001$
 (c) $\lim_{x \rightarrow 1^-} \frac{x+1}{x^3-1}$; $x = 0, 0.5, 0.9, 0.99, 0.999, 0.9999$
 14. (a) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x}$; $x = \pm 0.25, \pm 0.1, \pm 0.001, \pm 0.0001$
 (b) $\lim_{x \rightarrow 0^+} \frac{\sqrt{x+1}+1}{x}$; $x = 0.25, 0.1, 0.001, 0.0001$
 (c) $\lim_{x \rightarrow 0^-} \frac{\sqrt{x+1}+1}{x}$; $x = -0.25, -0.1, -0.001, -0.0001$
 15. (a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$; $x = \pm 0.25, \pm 0.1, \pm 0.001, \pm 0.0001$
 (b) $\lim_{x \rightarrow -1} \frac{\cos x}{x+1}$; $x = 0, -0.5, -0.9, -0.99, -0.999, -1.5, -1.1, -1.01, -1.001$
 16. (a) $\lim_{x \rightarrow -1} \frac{\tan(x+1)}{x+1}$; $x = 0, -0.5, -0.9, -0.99, -0.999, -1.5, -1.1, -1.01, -1.001$
 (b) $\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(2x)}$; $x = \pm 0.25, \pm 0.1, \pm 0.001, \pm 0.0001$

17–20 True–False Determine whether the statement is true or false. Explain your answer. ■

17. If $f(a) = L$, then $\lim_{x \rightarrow a} f(x) = L$.
 18. If $\lim_{x \rightarrow a} f(x)$ exists, then so do $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$.
 19. If $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exist, then so does $\lim_{x \rightarrow a} f(x)$.
 20. If $\lim_{x \rightarrow a^+} f(x) = +\infty$, then $f(a)$ is undefined.

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x)$						

▲ Table Ex-11

12. $f(x) = \frac{\cos x - 1}{x^2}$; $\lim_{x \rightarrow 0} f(x)$

x	-0.5	-0.05	-0.005	0.005	0.05	0.5
$f(x)$						

▲ Table Ex-12

- C 13–16** (i) Make a guess at the limit (if it exists) by evaluating the function at the specified x -values. (ii) Confirm your conclusions about the limit by graphing the function over an appropriate interval. (iii) If you have a CAS, then use it to find the limit. [Note: For the trigonometric functions, be sure to put your calculating and graphing utilities in radian mode.] ■

25. (i) $f(-3) = f(0) = f(2) = 0$
 (ii) $\lim_{x \rightarrow -2^-} f(x) = +\infty$ and $\lim_{x \rightarrow -2^+} f(x) = -\infty$
 (iii) $\lim_{x \rightarrow 1} f(x) = +\infty$
26. (i) $f(-1) = 0$, $f(0) = 1$, $f(1) = 0$
 (ii) $\lim_{x \rightarrow -1^-} f(x) = 0$ and $\lim_{x \rightarrow -1^+} f(x) = +\infty$
 (iii) $\lim_{x \rightarrow 1^-} f(x) = 1$ and $\lim_{x \rightarrow 1^+} f(x) = +\infty$

27–30 Modify the argument of Example 1 to find the equation of the tangent line to the specified graph at the point given. ■

27. the graph of $y = x^2$ at $(-1, 1)$
 28. the graph of $y = x^2$ at $(0, 0)$
 29. the graph of $y = x^4$ at $(1, 1)$
 30. the graph of $y = x^4$ at $(-1, 1)$

EXERCISE SET 1.2

1. Given that

$$\lim_{x \rightarrow a} f(x) = 2, \quad \lim_{x \rightarrow a} g(x) = -4, \quad \lim_{x \rightarrow a} h(x) = 0$$

find the limits.

(a) $\lim_{x \rightarrow a} [f(x) + 2g(x)]$

(b) $\lim_{x \rightarrow a} [h(x) - 3g(x) + 1]$

(c) $\lim_{x \rightarrow a} [f(x)g(x)]$ (d) $\lim_{x \rightarrow a} [g(x)]^2$

(e) $\lim_{x \rightarrow a} \sqrt[3]{6 + f(x)}$ (f) $\lim_{x \rightarrow a} \frac{2}{g(x)}$

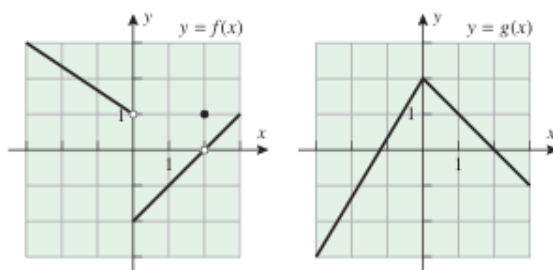
2. Use the graphs of f and g in the accompanying figure to find the limits that exist. If the limit does not exist, explain why.

(a) $\lim_{x \rightarrow 2} [f(x) + g(x)]$ (b) $\lim_{x \rightarrow 0} [f(x) + g(x)]$

(c) $\lim_{x \rightarrow 0^+} [f(x) + g(x)]$ (d) $\lim_{x \rightarrow 0^-} [f(x) + g(x)]$

(e) $\lim_{x \rightarrow 2} \frac{f(x)}{1 + g(x)}$ (f) $\lim_{x \rightarrow 2} \frac{1 + g(x)}{f(x)}$

(g) $\lim_{x \rightarrow 0^+} \sqrt{f(x)}$ (h) $\lim_{x \rightarrow 0^-} \sqrt{f(x)}$



▲ Figure Ex-2

3–30 Find the limits. ■

3. $\lim_{x \rightarrow 2} x(x-1)(x+1)$

5. $\lim_{x \rightarrow 3} \frac{x^2 - 2x}{x + 1}$

7. $\lim_{x \rightarrow 1^+} \frac{x^4 - 1}{x - 1}$

9. $\lim_{x \rightarrow -1} \frac{x^2 + 6x + 5}{x^2 - 3x - 4}$

11. $\lim_{x \rightarrow -1} \frac{2x^2 + x - 1}{x + 1}$

13. $\lim_{t \rightarrow 2} \frac{t^3 + 3t^2 - 12t + 4}{t^3 - 4t}$

15. $\lim_{x \rightarrow 3^+} \frac{x}{x - 3}$

17. $\lim_{x \rightarrow 3} \frac{x}{x - 3}$

19. $\lim_{x \rightarrow 2^-} \frac{x}{x^2 - 4}$

21. $\lim_{y \rightarrow 6^+} \frac{y + 6}{y^2 - 36}$

23. $\lim_{y \rightarrow 6} \frac{y + 6}{y^2 - 36}$

25. $\lim_{x \rightarrow 4^-} \frac{3 - x}{x^2 - 2x - 8}$

27. $\lim_{x \rightarrow 2^+} \frac{1}{|2 - x|}$

29. $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3}$

31. Let

$$f(x) = \begin{cases} x - 1, & x \leq 3 \\ 3x - 7, & x > 3 \end{cases}$$

4. $\lim_{x \rightarrow 3} x^3 - 3x^2 + 9x$

6. $\lim_{x \rightarrow 0} \frac{6x - 9}{x^3 - 12x + 3}$

8. $\lim_{t \rightarrow -2} \frac{t^3 + 8}{t + 2}$

10. $\lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^2 + x - 6}$

12. $\lim_{x \rightarrow 1} \frac{3x^2 - x - 2}{2x^2 + x - 3}$

14. $\lim_{t \rightarrow 1} \frac{t^3 + t^2 - 5t + 3}{t^3 - 3t + 2}$

16. $\lim_{x \rightarrow 3^-} \frac{x}{x - 3}$

18. $\lim_{x \rightarrow 2^+} \frac{x}{x^2 - 4}$

20. $\lim_{x \rightarrow 2} \frac{x}{x^2 - 4}$

22. $\lim_{y \rightarrow 6^-} \frac{y + 6}{y^2 - 36}$

24. $\lim_{x \rightarrow 4^+} \frac{3 - x}{x^2 - 2x - 8}$

26. $\lim_{x \rightarrow 4} \frac{3 - x}{x^2 - 2x - 8}$

28. $\lim_{x \rightarrow 3^-} \frac{1}{|x - 3|}$

30. $\lim_{y \rightarrow 4} \frac{4 - y}{2 - \sqrt{y}}$

Find

(a) $\lim_{x \rightarrow 3^-} f(x)$

(b) $\lim_{x \rightarrow 3^+} f(x)$

(c) $\lim_{x \rightarrow 3} f(x)$.

32. Let

$$g(t) = \begin{cases} t - 2, & t < 0 \\ t^2, & 0 \leq t \leq 2 \\ 2t, & t > 2 \end{cases}$$

Find

(a) $\lim_{t \rightarrow 0} g(t)$

(b) $\lim_{t \rightarrow 1} g(t)$

(c) $\lim_{t \rightarrow 2} g(t)$.

33–36 True–False Determine whether the statement is true or false. Explain your answer. ■

33. If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, then so does $\lim_{x \rightarrow a} [f(x) + g(x)]$.

34. If $\lim_{x \rightarrow a} g(x) = 0$ and $\lim_{x \rightarrow a} f(x)$ exists, then $\lim_{x \rightarrow a} [f(x)/g(x)]$ does not exist.

35. If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exist and are equal, then $\lim_{x \rightarrow a} [f(x)/g(x)] = 1$.

36. If $f(x)$ is a rational function and $x = a$ is in the domain of f , then $\lim_{x \rightarrow a} f(x) = f(a)$.

37–38 First rationalize the numerator and then find the limit. ■

37. $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$

38. $\lim_{x \rightarrow 0} \frac{\sqrt{x^2+4} - 2}{x}$

39. Let

$$f(x) = \frac{x^3 - 1}{x - 1}$$

(a) Find $\lim_{x \rightarrow 1} f(x)$.

(b) Sketch the graph of $y = f(x)$.

40. Let

$$f(x) = \begin{cases} \frac{x^2 - 9}{x + 3}, & x \neq -3 \\ k, & x = -3 \end{cases}$$

(a) Find k so that $f(-3) = \lim_{x \rightarrow -3} f(x)$.

(b) With k assigned the value $\lim_{x \rightarrow -3} f(x)$, show that $f(x)$ can be expressed as a polynomial.

Sarav

EXERCISE SET 1.3



1–4 In these exercises, make reasonable assumptions about the end behavior of the indicated function. ■

1. For the function g graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow -\infty} g(x)$ (b) $\lim_{x \rightarrow +\infty} g(x)$.

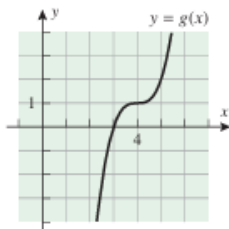


Figure Ex-1

2. For the function ϕ graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow -\infty} \phi(x)$
 (b) $\lim_{x \rightarrow +\infty} \phi(x)$.

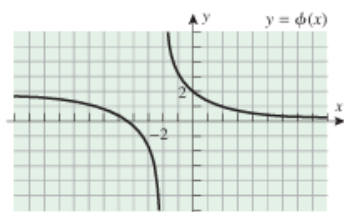


Figure Ex-2

3. For the function ϕ graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow -\infty} \phi(x)$ (b) $\lim_{x \rightarrow +\infty} \phi(x)$.

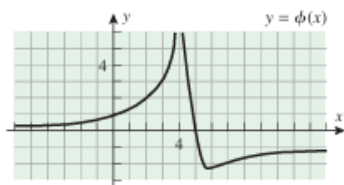


Figure Ex-3

4. For the function G graphed in the accompanying figure, find
 (a) $\lim_{x \rightarrow -\infty} G(x)$ (b) $\lim_{x \rightarrow +\infty} G(x)$.

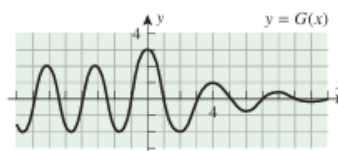


Figure Ex-4

5. Given that

$$\lim_{x \rightarrow +\infty} f(x) = 3, \quad \lim_{x \rightarrow +\infty} g(x) = -5, \quad \lim_{x \rightarrow +\infty} h(x) = 0$$

find the limits that exist. If the limit does not exist, explain why.

- (a) $\lim_{x \rightarrow +\infty} [f(x) + 3g(x)]$
 (b) $\lim_{x \rightarrow +\infty} [h(x) - 4g(x) + 1]$
 (c) $\lim_{x \rightarrow +\infty} [f(x)g(x)]$ (d) $\lim_{x \rightarrow +\infty} [g(x)]^2$
 (e) $\lim_{x \rightarrow +\infty} \sqrt[3]{5 + f(x)}$ (f) $\lim_{x \rightarrow +\infty} \frac{3}{g(x)}$
 (g) $\lim_{x \rightarrow +\infty} \frac{3h(x) + 4}{x^2}$ (h) $\lim_{x \rightarrow +\infty} \frac{6f(x)}{5f(x) + 3g(x)}$

6. Given that

$$\lim_{x \rightarrow -\infty} f(x) = 7 \quad \text{and} \quad \lim_{x \rightarrow -\infty} g(x) = -6$$

find the limits that exist. If the limit does not exist, explain why.

- (a) $\lim_{x \rightarrow -\infty} [2f(x) - g(x)]$ (b) $\lim_{x \rightarrow -\infty} [6f(x) + 7g(x)]$
 (c) $\lim_{x \rightarrow -\infty} [x^2 + g(x)]$ (d) $\lim_{x \rightarrow -\infty} [x^2 g(x)]$
 (e) $\lim_{x \rightarrow -\infty} \sqrt[3]{f(x)g(x)}$ (f) $\lim_{x \rightarrow -\infty} \frac{g(x)}{f(x)}$
 (g) $\lim_{x \rightarrow -\infty} \left[f(x) + \frac{g(x)}{x} \right]$ (h) $\lim_{x \rightarrow -\infty} \frac{xf(x)}{(2x + 3)g(x)}$

7. Complete the table and make a guess about the limit indicated.

$$f(x) = \frac{\sqrt{x^2 + x}}{x + 1} \quad \lim_{x \rightarrow +\infty} f(x)$$

x	10	100	1000	10,000	100,000	1,000,000
$f(x)$						

8. Complete the table and make a guess about the limit indicated.

$$f(x) = \frac{\sqrt{x^2 + x}}{x + 1} \quad \lim_{x \rightarrow -\infty} f(x)$$

x	-10	-100	-1000	-10,000	-100,000	-1,000,000
$f(x)$						

9–32 Find the limits. ■

9. $\lim_{x \rightarrow +\infty} (1 + 2x - 3x^5)$
10. $\lim_{x \rightarrow +\infty} (2x^3 - 100x + 5)$
11. $\lim_{x \rightarrow +\infty} \sqrt{x}$
12. $\lim_{x \rightarrow -\infty} \sqrt{5 - x}$
13. $\lim_{x \rightarrow +\infty} \frac{3x + 1}{2x - 5}$
14. $\lim_{x \rightarrow +\infty} \frac{5x^2 - 4x}{2x^2 + 3}$
15. $\lim_{y \rightarrow -\infty} \frac{3}{y + 4}$
16. $\lim_{x \rightarrow +\infty} \frac{1}{x - 12}$
17. $\lim_{x \rightarrow -\infty} \frac{x - 2}{x^2 + 2x + 1}$
18. $\lim_{x \rightarrow +\infty} \frac{5x^2 + 7}{3x^2 - x}$
19. $\lim_{x \rightarrow +\infty} \frac{7 - 6x^5}{x + 3}$
20. $\lim_{t \rightarrow -\infty} \frac{5 - 2t^3}{t^2 + 1}$
21. $\lim_{t \rightarrow +\infty} \frac{6 - t^3}{7t^3 + 3}$
22. $\lim_{x \rightarrow -\infty} \frac{x + 4x^3}{1 - x^2 + 7x^3}$
23. $\lim_{x \rightarrow +\infty} \sqrt[3]{\frac{2 + 3x - 5x^2}{1 + 8x^2}}$
24. $\lim_{s \rightarrow +\infty} \sqrt[3]{\frac{3s^7 - 4s^5}{2s^7 + 1}}$
25. $\lim_{x \rightarrow -\infty} \frac{\sqrt{5x^2 - 2}}{x + 3}$
26. $\lim_{x \rightarrow +\infty} \frac{\sqrt{5x^2 - 2}}{x + 3}$
27. $\lim_{y \rightarrow -\infty} \frac{2 - y}{\sqrt{7 + 6y^2}}$
28. $\lim_{y \rightarrow +\infty} \frac{2 - y}{\sqrt{7 + 6y^2}}$
29. $\lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8}$
30. $\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8}$
31. $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 3} - x)$
32. $\lim_{x \rightarrow +\infty} (\sqrt{x^2 - 3x} - x)$

33–36 True–False Determine whether the statement is true or false. Explain your answer. ■

33. By subtraction, $\lim_{x \rightarrow +\infty} (x^2 - 1000x) = \infty - \infty = 0$.
34. If $y = L$ is a horizontal asymptote for the curve $y = f(x)$, then

$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow +\infty} f(x) = L$$



35. If $y = L$ is a horizontal asymptote for the curve $y = f(x)$, then it is possible for the graph of f to intersect the line $y = L$ infinitely many times.
36. If a rational function $p(x)/q(x)$ has a horizontal asymptote, then the degree of $p(x)$ must equal the degree of $q(x)$.

39. Let

$$f(x) = \begin{cases} 2x^2 + 5, & x < 0 \\ \frac{3 - 5x^3}{1 + 4x + x^3}, & x \geq 0 \end{cases}$$

Find

(a) $\lim_{x \rightarrow -\infty} f(x)$

(b) $\lim_{x \rightarrow +\infty} f(x)$.

40. Let

$$g(t) = \begin{cases} \frac{2 + 3t}{5t^2 + 6}, & t < 1,000,000 \\ \frac{\sqrt{36t^2 - 100}}{5 - t}, & t > 1,000,000 \end{cases}$$

Find

(a) $\lim_{t \rightarrow -\infty} g(t)$

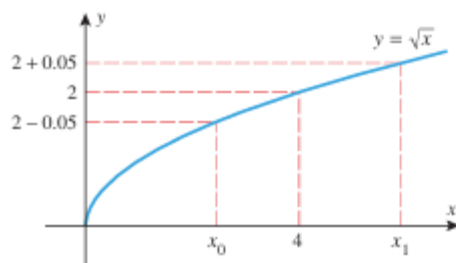
(b) $\lim_{t \rightarrow +\infty} g(t)$.

Sarah Nash

EXERCISE SET 1.4



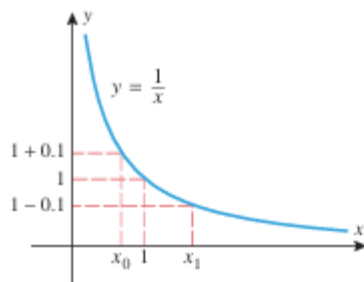
1. (a) Find the largest open interval, centered at the origin on the x -axis, such that for each x in the interval the value of the function $f(x) = x + 2$ is within 0.1 unit of the number $f(0) = 2$.
 (b) Find the largest open interval, centered at $x = 3$, such that for each x in the interval the value of the function $f(x) = 4x - 5$ is within 0.01 unit of the number $f(3) = 7$.
 (c) Find the largest open interval, centered at $x = 4$, such that for each x in the interval the value of the function $f(x) = x^2$ is within 0.001 unit of the number $f(4) = 16$.
2. In each part, find the largest open interval, centered at $x = 0$, such that for each x in the interval the value of $f(x) = 2x + 3$ is within ϵ units of the number $f(0) = 3$.
 (a) $\epsilon = 0.1$ (b) $\epsilon = 0.01$
 (c) $\epsilon = 0.0012$
3. (a) Find the values of x_0 and x_1 in the accompanying figure.
 (b) Find a positive number δ such that $|\sqrt{x} - 2| < 0.05$ if $0 < |x - 4| < \delta$.



Not drawn to scale

▲ Figure Ex-3

4. (a) Find the values of x_0 and x_1 in the accompanying figure.
 (b) Find a positive number δ such that $|(1/x) - 1| < 0.1$ if $0 < |x - 1| < \delta$.



Not drawn to scale

◀ Figure Ex-4

9–16 A positive number ϵ and the limit L of a function f at a are given. Find a number δ such that $|f(x) - L| < \epsilon$ if $0 < |x - a| < \delta$. ■

9. $\lim_{x \rightarrow 4} 2x = 8$; $\epsilon = 0.1$ 10. $\lim_{x \rightarrow 3} (5x - 2) = 13$; $\epsilon = 0.01$

11. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$; $\epsilon = 0.05$

12. $\lim_{x \rightarrow -1/2} \frac{4x^2 - 1}{2x + 1} = -2$; $\epsilon = 0.05$

13. $\lim_{x \rightarrow 2} x^3 = 8$; $\epsilon = 0.001$

14. $\lim_{x \rightarrow 4} \sqrt{x} = 2$; $\epsilon = 0.001$

15. $\lim_{x \rightarrow 5} \frac{1}{x} = \frac{1}{5}$; $\epsilon = 0.05$ 16. $\lim_{x \rightarrow 0} |x| = 0$; $\epsilon = 0.05$

17–26 Use Definition 1.4.1 to prove that the limit is correct. ■

17. $\lim_{x \rightarrow 2} 3 = 3$ 18. $\lim_{x \rightarrow 4} (x + 2) = 6$

19. $\lim_{x \rightarrow 5} 3x = 15$ 20. $\lim_{x \rightarrow -1} (7x + 5) = -2$

21. $\lim_{x \rightarrow 0} \frac{2x^2 + x}{x} = 1$ 22. $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3} = -6$

23. $\lim_{x \rightarrow 1} f(x) = 3$, where $f(x) = \begin{cases} x + 2, & x \neq 1 \\ 10, & x = 1 \end{cases}$

24. $\lim_{x \rightarrow 2} f(x) = 5$, where $f(x) = \begin{cases} 9 - 2x, & x \neq 2 \\ 49, & x = 2 \end{cases}$

25. $\lim_{x \rightarrow 0} |x| = 0$

26. $\lim_{x \rightarrow 2} f(x) = 5$, where $f(x) = \begin{cases} 9 - 2x, & x < 2 \\ 3x - 1, & x > 2 \end{cases}$

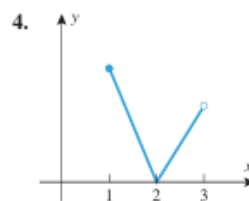
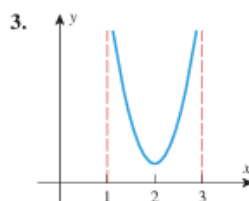
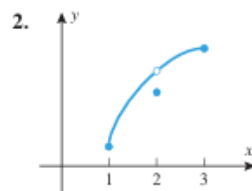
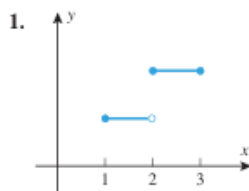
EXERCISE SET 1.5



1–4 Let f be the function whose graph is shown. On which of the following intervals, if any, is f continuous?

- (a) $[1, 3]$ (b) $(1, 3)$ (c) $[1, 2]$
(d) $(1, 2)$ (e) $[2, 3]$ (f) $(2, 3)$

For each interval on which f is not continuous, indicate which conditions for the continuity of f do not hold. ■



5. Consider the functions

$$f(x) = \begin{cases} 1, & x \neq 4 \\ -1, & x = 4 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 4x - 10, & x \neq 4 \\ -6, & x = 4 \end{cases}$$

In each part, is the given function continuous at $x = 4$?

- (a) $f(x)$ (b) $g(x)$ (c) $-g(x)$ (d) $|f(x)|$
(e) $f(x)g(x)$ (f) $g(f(x))$ (g) $g(x) - 6f(x)$



6. Consider the functions

$$f(x) = \begin{cases} 1, & 0 \leq x \\ 0, & x < 0 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 0, & 0 \leq x \\ 1, & x < 0 \end{cases}$$

In each part, is the given function continuous at $x = 0$?

- (a) $f(x)$ (b) $g(x)$ (c) $f(-x)$ (d) $|g(x)|$
(e) $f(x)g(x)$ (f) $g(f(x))$ (g) $f(x) + g(x)$



11–22 Find values of x , if any, at which f is not continuous. ■

11. $f(x) = 5x^4 - 3x + 7$ 12. $f(x) = \sqrt[3]{x - 8}$

13. $f(x) = \frac{x+2}{x^2+4}$ 14. $f(x) = \frac{x+2}{x^2-4}$

15. $f(x) = \frac{x}{2x^2+x}$ 16. $f(x) = \frac{2x+1}{4x^2+4x+5}$

17. $f(x) = \frac{3}{x} + \frac{x-1}{x^2-1}$ 18. $f(x) = \frac{5}{x} + \frac{2x}{x+4}$

19. $f(x) = \frac{x^2+6x+9}{|x|+3}$ 20. $f(x) = \left| 4 - \frac{8}{x^4+x} \right|$

21. $f(x) = \begin{cases} 2x+3, & x \leq 4 \\ 7 + \frac{16}{x}, & x > 4 \end{cases}$

22. $f(x) = \begin{cases} \frac{3}{x-1}, & x \neq 1 \\ 3, & x = 1 \end{cases}$

23–28 True–False Determine whether the statement is true or false. Explain your answer. ■

- 23. If $f(x)$ is continuous at $x = c$, then so is $|f(x)|$.
- 24. If $|f(x)|$ is continuous at $x = c$, then so is $f(x)$.
- 25. If f and g are discontinuous at $x = c$, then so is $f + g$.
- 26. If f and g are discontinuous at $x = c$, then so is fg .
- 27. If $\sqrt{f(x)}$ is continuous at $x = c$, then so is $f(x)$.
- 28. If $f(x)$ is continuous at $x = c$, then so is $\sqrt{f(x)}$.

29–30 Find a value of the constant k , if possible, that will make the function continuous everywhere. ■

- 29. (a) $f(x) = \begin{cases} 7x - 2, & x \leq 1 \\ kx^2, & x > 1 \end{cases}$
(b) $f(x) = \begin{cases} kx^2, & x \leq 2 \\ 2x + k, & x > 2 \end{cases}$
- 30. (a) $f(x) = \begin{cases} 9 - x^2, & x \geq -3 \\ k/x^2, & x < -3 \end{cases}$
(b) $f(x) = \begin{cases} 9 - x^2, & x \geq 0 \\ k/x^2, & x < 0 \end{cases}$

31. Find values of the constants k and m , if possible, that will make the function f continuous everywhere.

$$f(x) = \begin{cases} x^2 + 5, & x > 2 \\ m(x + 1) + k, & -1 < x \leq 2 \\ 2x^3 + x + 7, & x \leq -1 \end{cases}$$

32. On which of the following intervals is

$$f(x) = \frac{1}{\sqrt{x-2}}$$

continuous?

- (a) $[2, +\infty)$ (b) $(-\infty, +\infty)$ (c) $(2, +\infty)$ (d) $[1, 2)$

33–36 A function f is said to have a **removable discontinuity** at $x = c$ if $\lim_{x \rightarrow c} f(x)$ exists but f is not continuous at $x = c$, either because f is not defined at c or because the definition for $f(c)$ differs from the value of the limit. This terminology will be needed in these exercises. ■

- 33.** (a) Sketch the graph of a function with a removable discontinuity at $x = c$ for which $f(c)$ is undefined.
 (b) Sketch the graph of a function with a removable discontinuity at $x = c$ for which $f(c)$ is defined.
- 34.** (a) The terminology *removable discontinuity* is appropriate because a removable discontinuity of a function f at $x = c$ can be “removed” by redefining the value of f appropriately at $x = c$. What value for $f(c)$ removes the discontinuity?
 (b) Show that the following functions have removable discontinuities at $x = 1$, and sketch their graphs.

$$f(x) = \frac{x^2 - 1}{x - 1} \quad \text{and} \quad g(x) = \begin{cases} 1, & x > 1 \\ 0, & x = 1 \\ 1, & x < 1 \end{cases}$$

- (c) What values should be assigned to $f(1)$ and $g(1)$ to remove the discontinuities?

35–36 Find the values of x (if any) at which f is not continuous, and determine whether each such value is a removable discontinuity. ■

35. (a) $f(x) = \frac{|x|}{x}$

(b) $f(x) = \frac{x^2 + 3x}{x + 3}$

(c) $f(x) = \frac{x - 2}{|x| - 2}$

36. (a) $f(x) = \frac{x^2 - 4}{x^3 - 8}$

(b) $f(x) = \begin{cases} 2x - 3, & x \leq 2 \\ x^2, & x > 2 \end{cases}$

(c) $f(x) = \begin{cases} 3x^2 + 5, & x \neq 1 \\ 6, & x = 1 \end{cases}$

EXERCISE SET 1.6

Graphing Utility

1–8 Find the discontinuities, if any. ■

1. $f(x) = \sin(x^2 - 2)$

2. $f(x) = \cos\left(\frac{x}{x - \pi}\right)$

3. $f(x) = |\cot x|$

4. $f(x) = \sec x$

5. $f(x) = \csc x$

6. $f(x) = \frac{1}{1 + \sin^2 x}$

7. $f(x) = \frac{1}{1 - 2 \sin x}$

8. $f(x) = \sqrt{2 + \tan^2 x}$

11–32 Find the limits. ■

11. $\lim_{x \rightarrow +\infty} \cos\left(\frac{1}{x}\right)$

12. $\lim_{x \rightarrow +\infty} \sin\left(\frac{\pi x}{2 - 3x}\right)$

13. $\lim_{\theta \rightarrow 0} \frac{\sin 3\theta}{\theta}$

14. $\lim_{h \rightarrow 0} \frac{\sin h}{2h}$

15. $\lim_{x \rightarrow 0} \frac{x^2 - 3 \sin x}{x}$

16. $\lim_{x \rightarrow 0} \frac{2 - \cos 3x - \cos 4x}{x}$

17. $\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta^2}$

18. $\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta}$

19. $\lim_{x \rightarrow 0} \frac{\tan 7x}{\sin 3x}$

20. $\lim_{x \rightarrow 0} \frac{\sin 6x}{\sin 8x}$

21. $\lim_{x \rightarrow 0^+} \frac{\sin x}{5\sqrt{x}}$

22. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{3x^2}$

23. $\lim_{x \rightarrow 0} \frac{\sin x^2}{x}$

24. $\lim_{h \rightarrow 0} \frac{\sin h}{1 - \cos h}$

25. $\lim_{t \rightarrow 0} \frac{t^2}{1 - \cos^2 t}$

26. $\lim_{x \rightarrow 0} \frac{x}{\cos\left(\frac{1}{2}\pi - x\right)}$

27. $\lim_{\theta \rightarrow 0} \frac{\theta^2}{1 - \cos \theta}$

28. $\lim_{h \rightarrow 0} \frac{1 - \cos 3h}{\cos^2 5h - 1}$

29. $\lim_{x \rightarrow 0^+} \sin\left(\frac{1}{x}\right)$

30. $\lim_{x \rightarrow 0} \frac{\tan 3x^2 + \sin^2 5x}{x^2}$

31. $\lim_{x \rightarrow 0} \frac{\tan ax}{\sin bx}, \quad (a \neq 0, b \neq 0)$

32. $\lim_{x \rightarrow 0} \frac{\sin^2(kx)}{x}, \quad k \neq 0$